

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2018

Mathematics

Paper 1 Ordinary Level

Time: 2 hours

300 marks

School Stamp

Running Total	
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For Examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if you do not show all necessary work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1

(Suggested maximum time: 10 minutes)

12	46	8	20	76	13	33	56	88	15
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- (a) From the list of numbers above,
- (i) write down two numbers that have a **sum** of 45,

Answer = and .

[illegible]

- (ii) write down two numbers that have a **difference** of 30,

Answer = and

[illegible]

- (iii) write down a number that is a **multiple** of 7, Answer =

Answer =

[illegible]

- (iv)** write down a number that is a **factor** of 80,

Answer =

[illegible]

- (v) write down a **prime number**.

Answer =

- (b)** What is the **largest** number you can get by multiplying any two numbers from the list above?

Answer =

[illegible]

Question 2

(Suggested maximum time: 10 minutes)

- (a) Conor, Ray and Amy sat a History examination.
The total for the examination was 180 marks.
- (i) Conor got 117 marks out of the 180 marks.
What percentage mark did Conor get?

[illegible]

- (ii) Ray got $\frac{3}{4}$ of the 180 marks. How many marks did Ray get?

[illegible]

- (iii)** Amy got 0.8 of the 180 marks. How many marks did Amy get?

[illegible]

- (iv)** Who got the highest marks in the examination?

Answer =

- (b)** The amount of time Conor, Ray and Amy spent studying for the examination is shown in the table below. Complete the table.

	Total in Minutes	Total in Hours
Conor	30	$\frac{1}{2}$
Ray	40	
Amy		$1\frac{1}{4}$

[illegible]

(Suggested maximum time: 5 minutes)

(a) Factorise $x^2 - 49$. One of the factors is $(x + 7)$.

$$x^2 - 49 = (x + 7) (\quad)$$

(b) Using your factors from part **(a)**, or otherwise, solve the equation

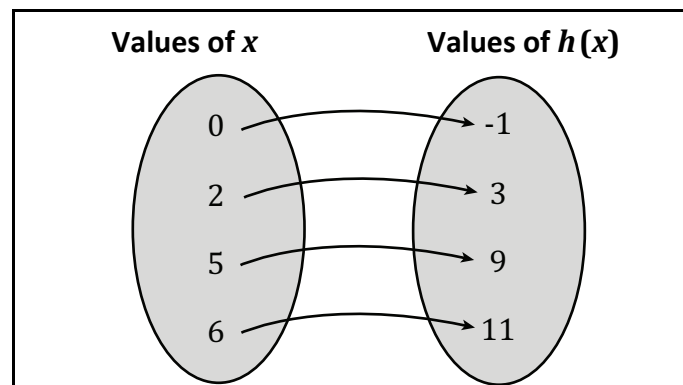
$$x^2 - 49 = 0.$$

[illegible]

(Suggested maximum time: 5 minutes)

A function h is defined as $h(x) = 2x - 1$.

The arrow diagram below shows the values of $h(x)$ for the given values of x .



(a) Write down the domain and the range of the function h .

Domain = { , , , } Range = { , , , }

(b) (i) Find the value of $h(4)$.

[illegible]

$$h(4) = \boxed{}$$

(ii) Find the value of $h(-3)$.

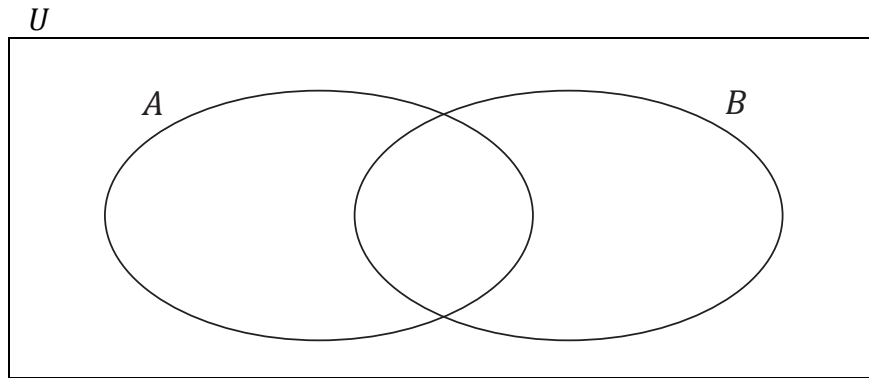
[illegible]

$$h(-3) = \boxed{}$$

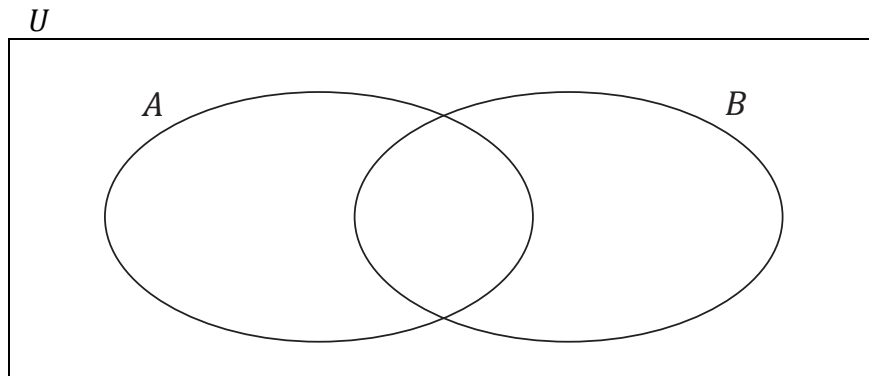
Question 5

(Suggested maximum time: 10 minutes)

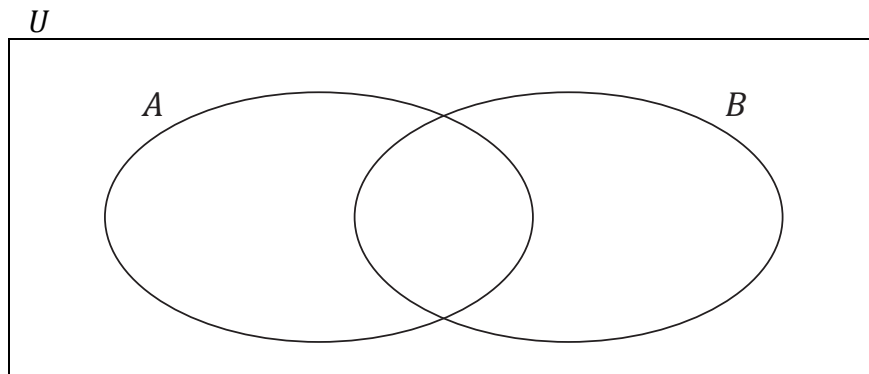
- (a) (i)** On the Venn diagram below, **shade in** the region that represents B .



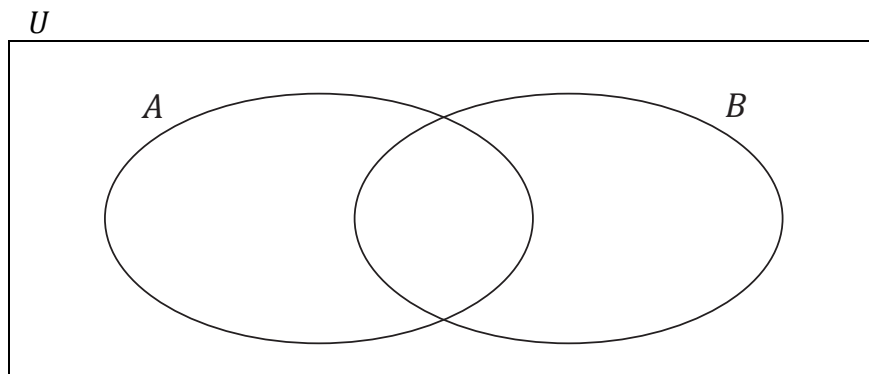
- (ii)** On the Venn diagram below, **shade in** the region that represents $A \cap B$.



- (iii)** On the Venn diagram below, **shade in** the region that represents $(A \cup B)'$, where $(A \cup B)'$ is the complement of $A \cup B$.



- (iv)** On the Venn diagram below, **shade in** the region that represents $A \setminus B$.



- 16 students like apples
9 students like bananas
5 students like apples and bananas.

-
- A Venn diagram illustrating the relationship between two sets, A and B , within a universal set U . The universal set U is represented by a large rectangle. Inside this rectangle, there are two overlapping ellipses. The left ellipse is labeled A and the right ellipse is labeled B . The two ellipses overlap in the center, representing the intersection of sets A and B .

[illegible]

- [illegible]

- | Statement | Tick one only for each statement. | |
|---------------------------------|--|-------|
| | True | False |
| $A \cup B = B \cup A$ | | |
| $A \setminus B = B \setminus A$ | | |
| $A \setminus B \subset A$ | | |
| $\#A = \#B$ | | |
| $U = A \cup B$ | | |

[illegible]

Question 6 (Suggested maximum time: 10 minutes)

Patrick, Debbie and Tom counted the number of text messages they received that day.



- (a) Patrick received x number of text messages.
Debbie received 4 more than Patrick. Tom received twice as many as Patrick.

Select suitable expressions from A , B , C , D and E below which represent the number of text messages that Debbie and Tom each received.

A	B	C	D	E
$4x$	$x + 4$	$x + 2$	$2x + 4$	$2x$

Debbie received	text messages
-----------------	---------------

Tom received		text messages
--------------	--	---------------

- (b)** Patrick said that the total number of text messages Debbie and Tom received between them can be represented by the expression $3x + 4$.
Show that he is correct.

[illegible]

- (c) The total number of text messages that Debbie and Tom received was 19. Write down an equation in x to represent this statement.

[illegible]

- (d)** Solve your equation to find the number of text messages that Patrick received.

[illegible]

- (e)** How many text messages did Debbie and Tom each receive?

Debbie:	Tom:
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Question 7

(Suggested maximum time: 10 minutes)

A shop sells hot chocolate in jars. Each jar contains 288 g of hot chocolate.

Anna makes 2 cups of hot chocolate each day.

She estimates that she uses 20 g of hot chocolate to make each cup.

Anna wants to buy enough hot chocolate for 21 days.



- (a)** How many jars of hot chocolate does Anna need to buy?

[illegible]

- (b)** Anna finds that she actually uses 23 g of hot chocolate to make each cup. How does this affect the number of jars of hot chocolate she needs to buy? Give a reason for your answer.

[illegible]

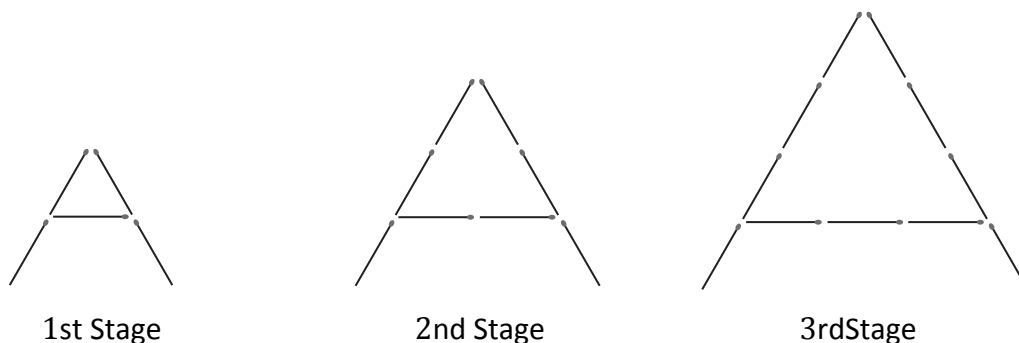
- (c) Another shop sells the same brand of hot chocolate in 1 kg tubs. A 288 g jar costs €4.15 and a 1 kg tub costs €15.95. Find the **cheapest** option available to Anna. Justify your answer fully. Remember that there are 1000 g in one kg.

A large grid of graph paper with 20 columns and 15 rows. The grid is composed of small squares, with a slightly larger square at the top left corner, likely for a title or header. The grid is empty and ready for use.

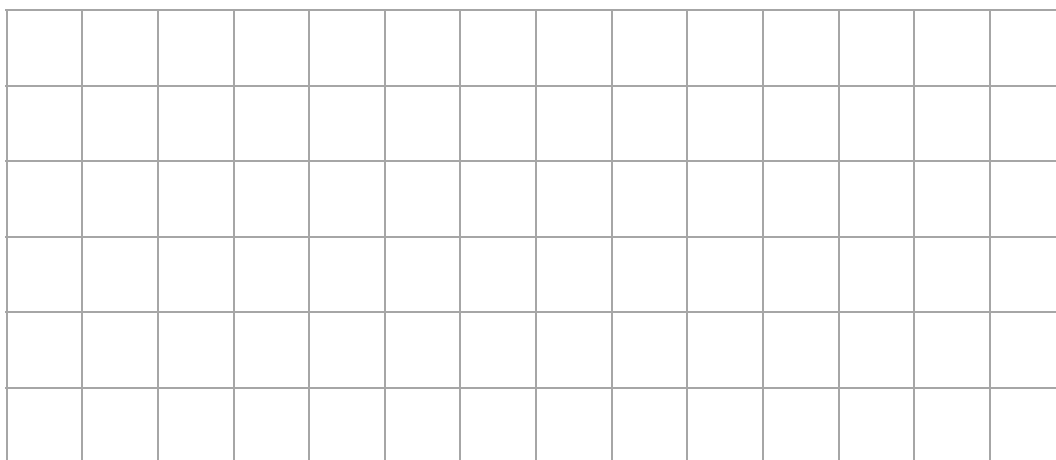
Question 8

(Suggested maximum time: 10 minutes)

The first three stages of a pattern are shown below.
Each stage is made up of a certain number of match sticks.



- (a) Draw the 4th stage of the pattern in the grid below.



- (b) Complete the table below to show the number of match sticks required for each of the first six stages of the pattern.

Stage	1st	2nd	3rd	4th	5th	6th
Number of match sticks						

- (c) What kind of sequence is made by the number of match sticks that are required for each stage of the pattern? Give a reason for your answer.

Kind of sequence: Linear Quadratic Exponential
(Tick (✓) **one** box only) ☐ ☐ ☐

Reason:

- (d) Find the number of match sticks required for the 14th stage of the pattern.

Question 9 (Suggested maximum time: 10 minutes)

Question 9 (Suggested maximum time: 10 minutes)

- (a)** Three expressions are shown below.

$x - y$

xy

$$\frac{x}{y}$$

Which expression has the smallest value when $x = 6$ and $y = 2$?

Answer =

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Give a reason for your answer.

[illegible]

- (b)** Solve the following simultaneous equations.

$$2x + 4y = 26$$

$$3x - y = 4$$

[illegible]

Question 10 (Suggested maximum time: 10 minutes)

Rob and Chloe are planning the meal for their wedding. Two local hotels offer the following prices:



Grand Hotel	
Number of people	Price per person
100 & over	€39.50
80 to 99	€41.50
60 to 79	€43.50
Up to 59	€45.50

- (a)** Find the cost of a meal in the Central Hotel with the special offer discount.

[illegible]

- (b)** Find the total cost of 63 meals in the Grand Hotel.

[illegible]

- (c)** Rob and Chloe have €4000 to spend on meals at their wedding.

What is the **maximum** number of meals they can pay for?

Answer =

Give a reason for your answer, comparing the maximum number of meals for both hotels.

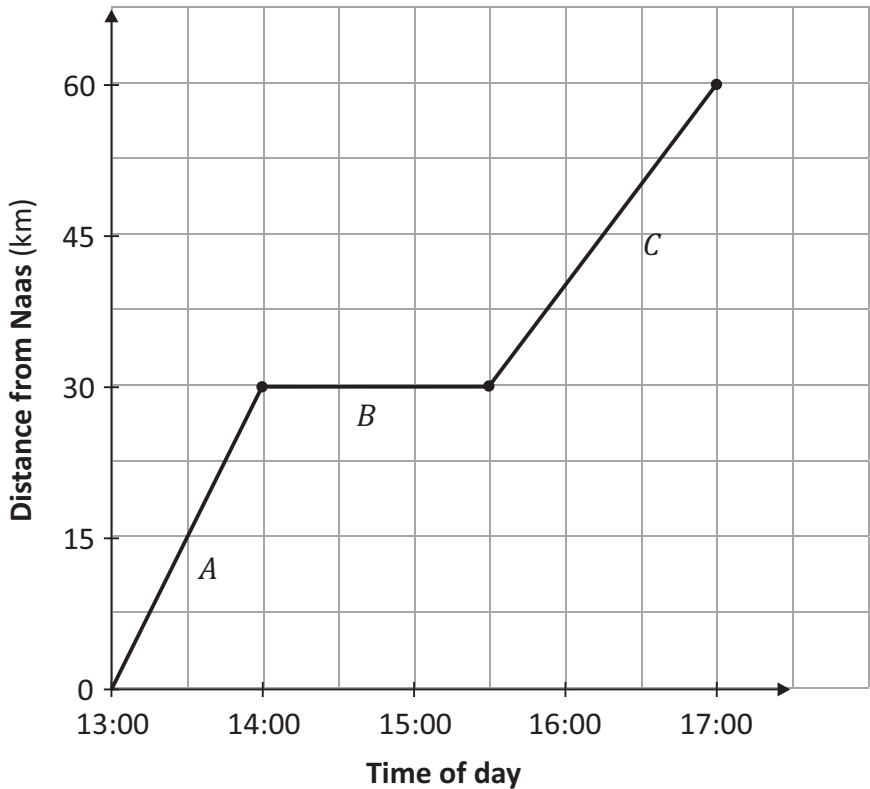
Reason:

[illegible]

Question 11

(Suggested maximum time: 5 minutes)

Sam drove 60 km, from Naas to Portlaoise.
He stopped and visited his friend Katie on the way.
The graph below shows his journey in three sections, labelled *A*, *B* and *C*.

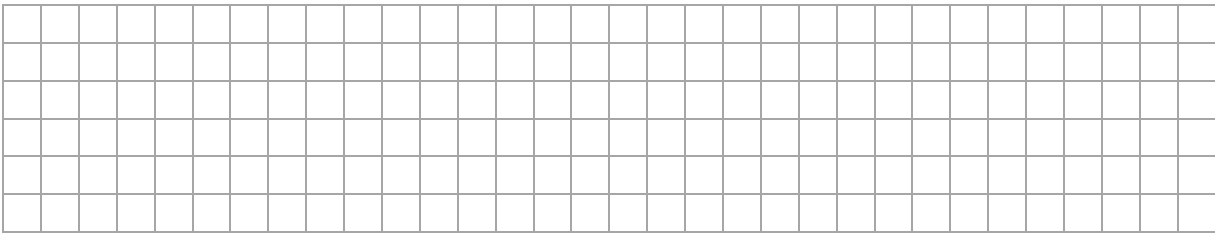


(a) What time did Sam arrive in Portlaoise?

Answer =

(b) What was Sam’s average speed for section *A* of the journey?

Answer =

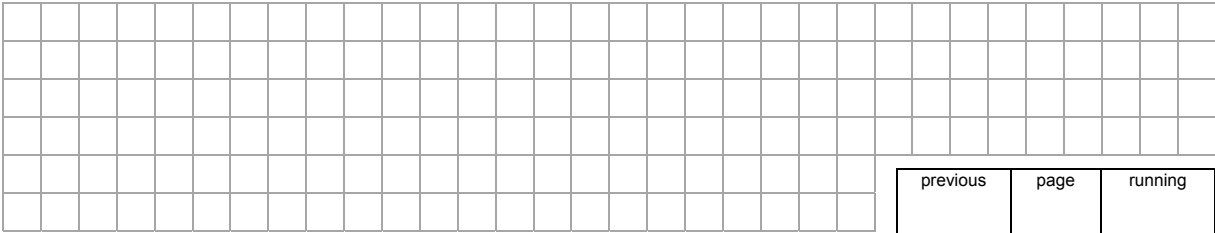


(c) How long did Sam spend visiting Katie?

Answer =

(d) What was Sam’s average speed for section *C* of the journey?

Answer =

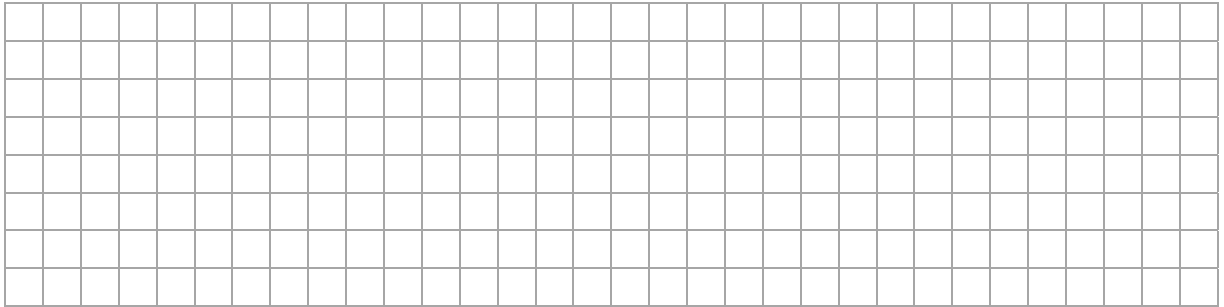


Question 12**(Suggested maximum time: 15 minutes)**

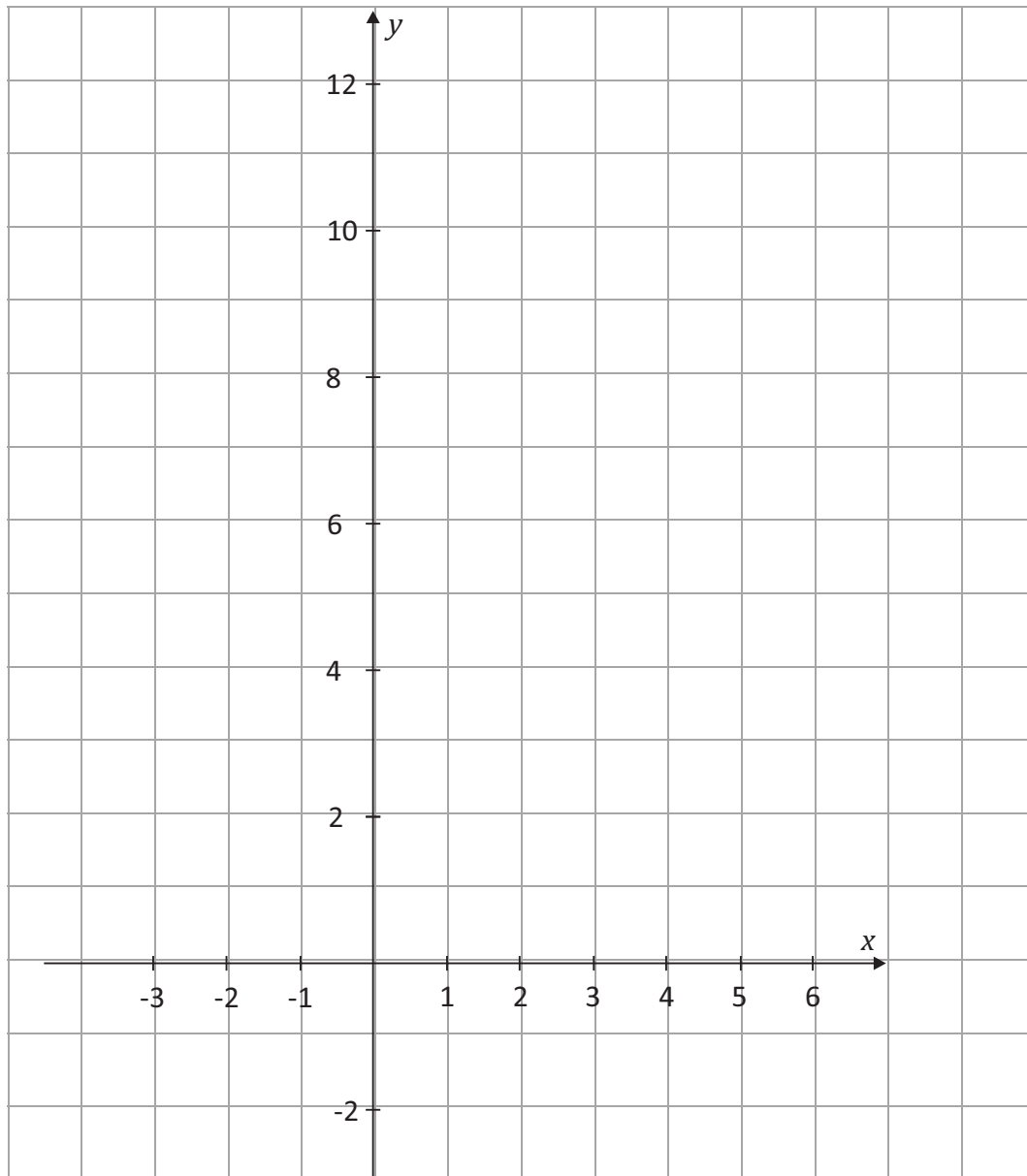
- (a) Carla says that $\{(-1, 1), (2, 4), (3, 5), (5, 7)\}$ are four couples of the function

$$f(x) = x + 2.$$

- (i) Show that Carla is correct.



- (ii) Using the four couples, draw the graph of the line $f(x) = x + 2$ for $-1 \leq x \leq 5$, $x \in \mathbb{R}$.



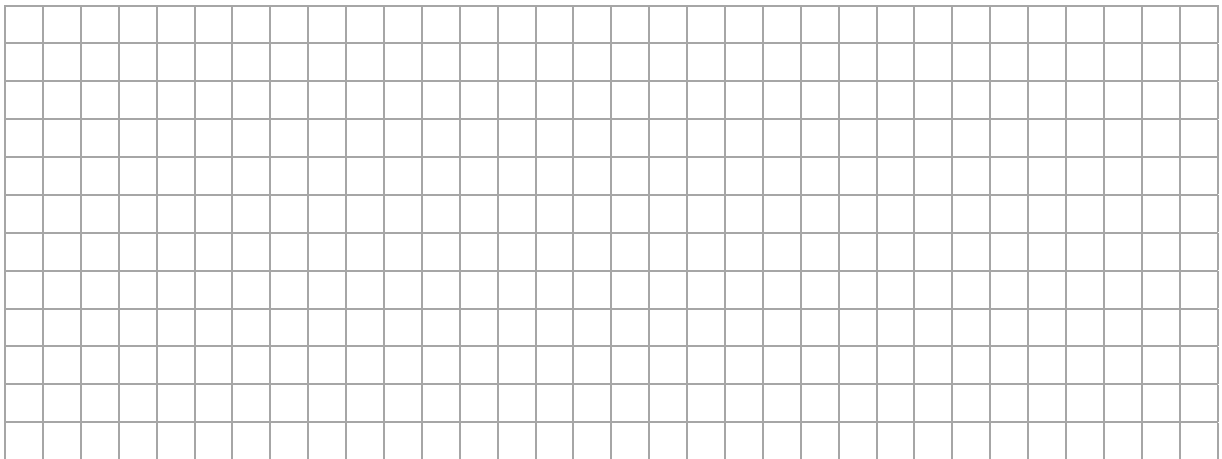
(b) Another function is $g : x \mapsto x^2 - 4x + 6$.

(i) Complete the table for the function

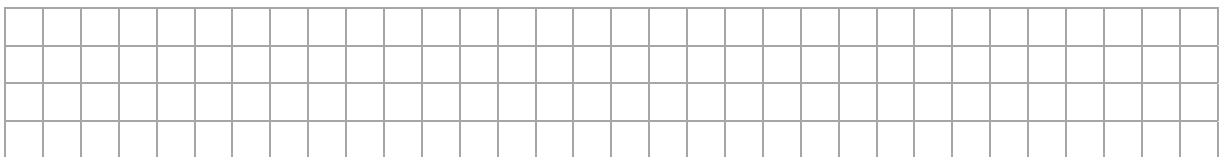
$$g : x \mapsto x^2 - 4x + 6$$

for $-1 \leq x \leq 5$, where $x \in \mathbb{R}$.

x	$g(x)$
-1	11
0	
1	
2	2
3	
4	
5	

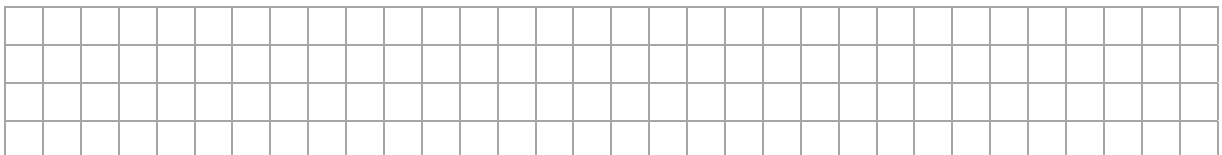


(ii) Using the values in the table, draw the graph of the function $g : x \mapsto x^2 - 4x + 6$ on the same axes used for part **(a)(ii)** for $-1 \leq x \leq 5$, $x \in \mathbb{R}$.

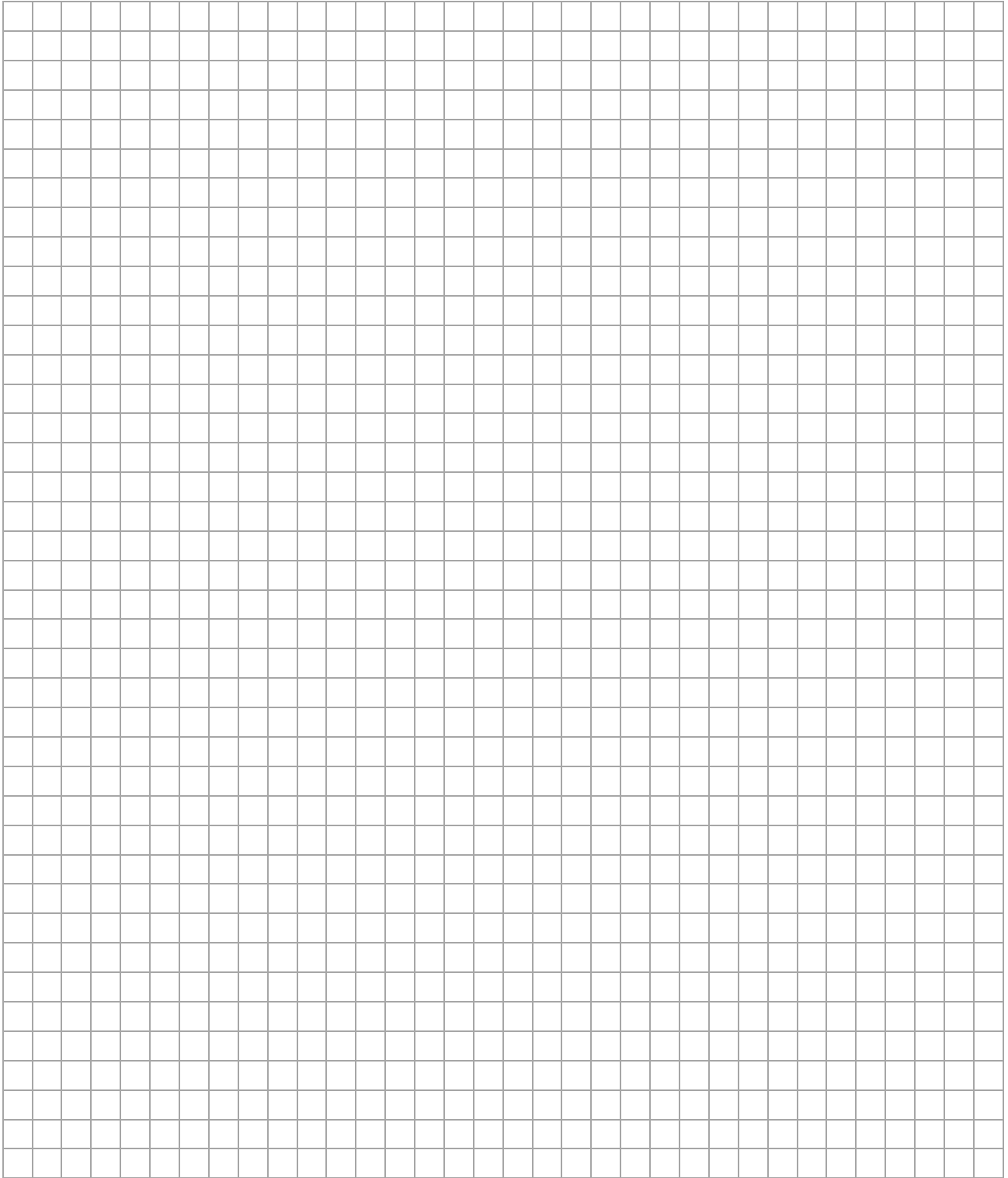


(iii) Use the graphs to write down the points of intersection of $f(x)$ and $g(x)$.

Answer = $\left(\quad , \quad \right)$ and $\left(\quad , \quad \right)$



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Pre-Junior Certificate, 2018 – Ordinary Level

Mathematics – Paper 1

Time: 2 hours

